



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE

Course Code & Title: CS3353 & C Programming & Data structures

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit II / Pointers**
- **Course Outcome: CO2**
- **Topic Learning Outcome: TLO4**
- **Activity Chosen: Crossword Puzzle**
- **Justification:**

Pointer is the important concept in C programming. Different types of pointers and the way in which representation of the pointer and address of the pointers are declared are easily paired with the help of crossword puzzle which leads to the greater clarity about the concept of pointer to the students.

- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
 - A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. Instead of administering standard MCQ tests, it sparks pupils' curiosity.
 - The activity was conducted in a classroom for a duration of 10 minutes. The students were given 10 questions as a crossword puzzle in the topic of pointers. They have to think and write the answers in the given puzzle.
 - The activity was conducted during the last 10 minutes of the class after that the answers were corrected. Sample answers were shown in figure 1. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.
- **CO – PO / PSO mapping:**

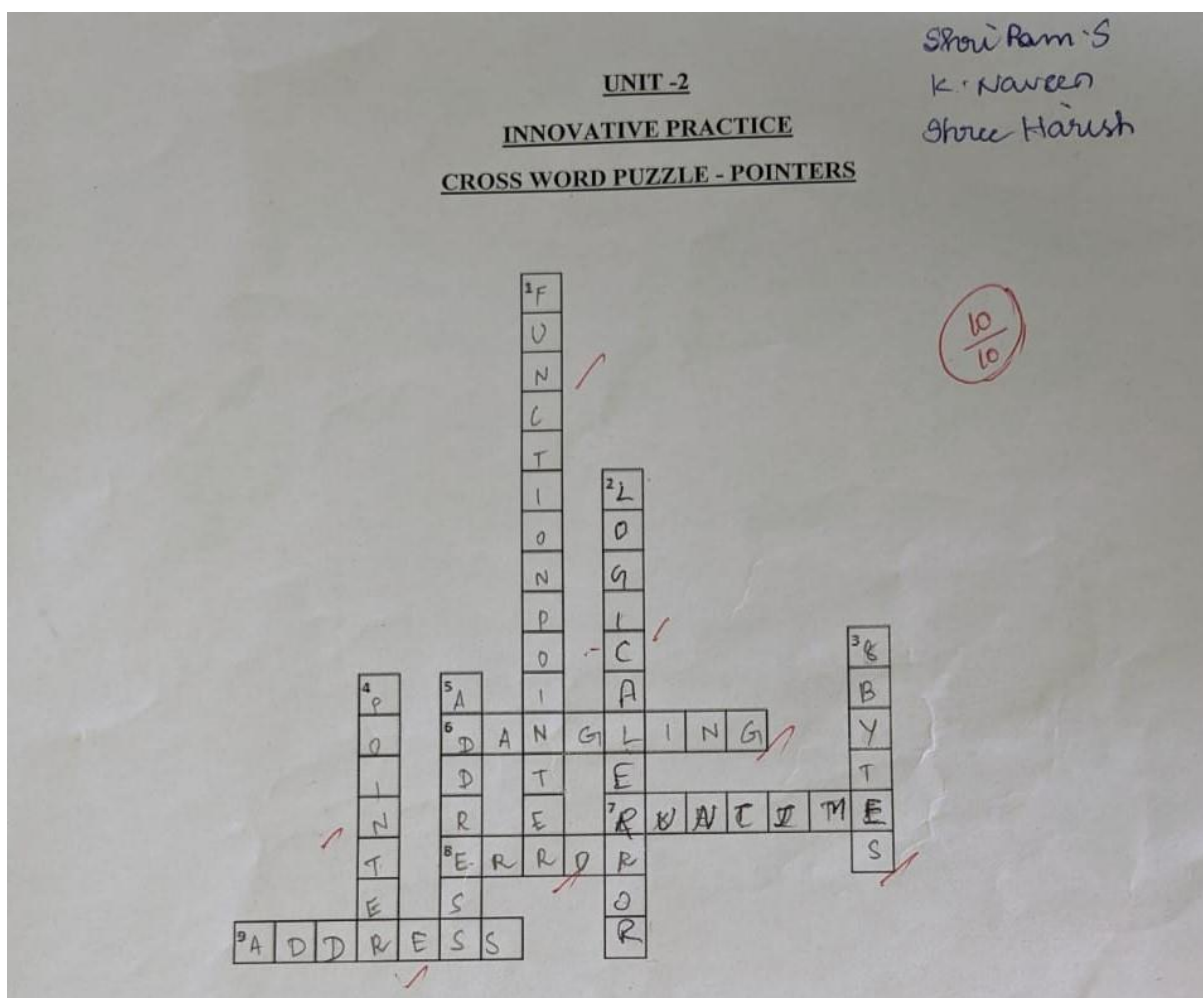
CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO1
CO2	3	2	1	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

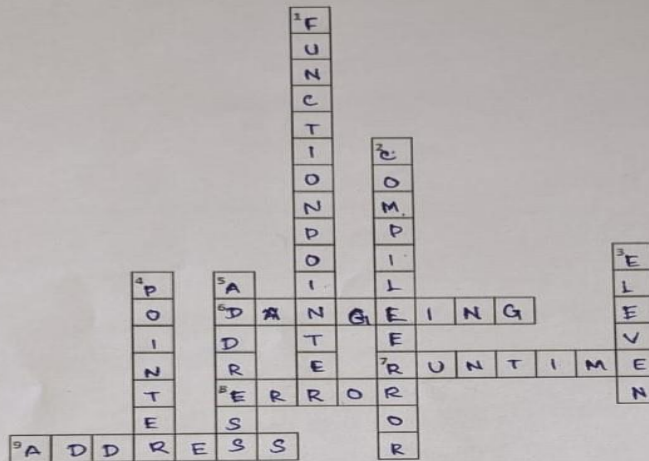
Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	1	2	1
Justification for correlation	The students apply the knowledge of pointers	The students analyze the given questions	The students find the solutions for the given question.	The students work in a team to find solutions for the given question.	The students communicate with their team members and explain the reason for providing particular solution for the question.	The students apply the knowledge of pointer to solve the question.

• **Images / Screenshot of the practice:**



K. Siva Kiruthika
K. S. Shobha

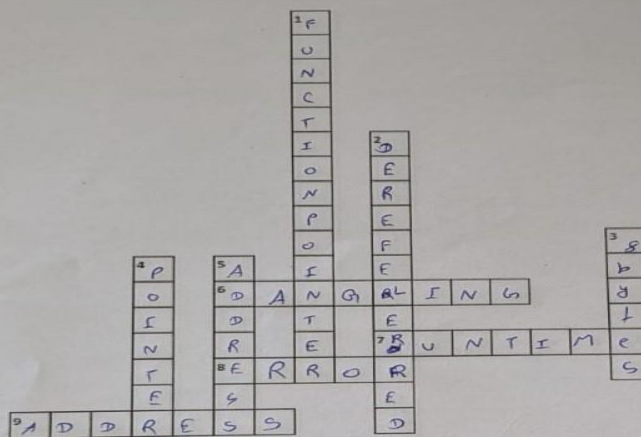
UNIT-2
INNOVATIVE PRACTICE
CROSS WORD PUZZLE - POINTERS



8/10

A. Santhosh Kumar
Saswathi vel.

UNIT-2
INNOVATIVE PRACTICE
CROSS WORD PUZZLE - POINTERS



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Across

6. dangling ~~delete~~ pointer arise when an object is deleted or de allocated, without modifying the value of the pointer.
7. Memory can be allocated dynamically during run time of the program.
8. Subtracting two pointer variable will create an error ~~address~~
9. A pointer is a variable that stores data ~~address~~ of another variable.

Down

1. function pointer ~~is~~ is a special pointer that points to a function.
2. Dereference a null pointer can create a dereferenced ~~error~~
3. Size of Character Pointer is 8 bytes
4. The operator * signifies the pointer
5. The & symbol is used to represent the address of the variable.

Fig 1: Sample Answers

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Some of the students find very easy to identify the answer for the given questions.
- Some of them take more time to complete the puzzle need more clarification on the various classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to think on their own and identify the use of pointers for the variables and access the value of the variable by passing the address of the variable. The students level of understanding on this topic gets increased.

- ❖ ***Challenges faced in implementation:***

- The students who didn't understand the concept clearly were unable to complete it within the allotted time.
- Few of them were struggled to find the answer for the questions.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ <https://crosswordlabs.com/>
- ❖ Reema Thareja, "Programming in C", Second Edition, Oxford University Press, 2016.

Signature of Faculty Member

HOD